Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)
Department of Computer Science and Engineering

CAT Assignment 2

Degree & Branch	M. Tech (Integrated) Computer Science & Engineering	Semester	V
Subject Code & Name	ICS1502 & Introduction to Machine Learning		
Academic year	2026-2027 (Odd)	Batch:2024-2029	**Due date:14-10-2026**

- **Instructions:** You are expected to complete the assignment independently. While you are free to engage in discussions with your peers, do not copy from one another. If we identify submissions from two or more students that are significantly similar in content, all students involved will receive zero marks for this assignment. All submissions will be checked for plagiarism using standard tools. Please adhere to the following submission guidelines:
  - Use Python for all coding tasks.
  - Submit your code in the form of a Jupyter Notebook (.ipynb).
  - Submit your answers to the theory questions in a separate document (PDF or Word format).

1 Programming Assignment

- Fit a Linear Classification Model or any single-layered or multi-layered neural network (2-class classification model) using the given dataset: Bank Note Authentication **dataset**. Critically evaluate and compare the suitability and performance of the chosen model in solving this binary classification problem by analyzing accuracy, generalization ability, robustness to outliers, and interpretability. Further, justify whether the linear model or the neural network architecture provides a more effective solution, and support your conclusion with appropriate metrics and reasoning. [CO3, CO4, CO5, K5, 2.3.1, 3.4.3, 5.1.2, 6.2.1, 8.2.1, 10.1.1, 10.3.1, 10.3.2, 12.2.2]

  1. **Linear Model Baseline:** Fit a linear classification model (e.g., Logistic Regression) with and without l_2 regularization and compare the training and test accuracies. (**5 marks**) [CO3, K5, 10.3.1, 10.3.2]
  2. **Regularization Effect:** Plot training and test accuracy as a function of λ and critically evaluate the effect of regularization strength on overfitting and generalization. (**5 marks**) [CO3, K5, 8.2.1]
  3. **Decision Boundary Visualization:** Visualize decision boundaries in a 3D feature space using three selected features and Interpret the extent to which the model achieves class separability. (**5 marks**) [CO3, K5, 3.4.3]
  4. **Neural Network Comparison:** Fit a single-layer or multi-layer neural network classifier on the same dataset. Compare its accuracy, precision, recall, F1-score, and generalization performance against the linear model. (**5 marks**) [CO4, K5, 2.2.4, 10.1.1]
  5. **Dimensionality Reduction:** Apply PCA for feature reduction and critically evaluate the classification performance in the reduced space compared with the original space. Interpret whether PCA improves model effectiveness. (**10 marks**) [CO4, K5, 5.1.2, 10.3.1]

6. **Statistical Significance Testing:** Apply the Friedman Test and One-Way ANOVA to test whether observed differences in performance among models are statistically significant. Critically evaluate the results, identify and justify which models perform significantly better, and interpret the practical implications for real-world deployment. (**10 marks**) [CO5, K5, 13.3.1, 14.7.1, 14.7.2]

Prepared by	**Reviewed by**	**Approved by**
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